

Contingencies and Implications of Artificial Consciousness: Can We Code Consciousness?

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Abstract—The problematic scheme of studying artificial consciousness is framed by the notion that there fails to exist a concise and agreed-upon definition of consciousness and its origins in humans. However, that does not diminish the need for probing the notion that non-human entities may one day be sentient. As the future of artificial consciousness remains both undetermined and controversial, it remains a worthwhile challenge to look into as we ponder the cognitive capabilities of machines. This paper outlines the current state of artificial consciousness and some of the current research that has been done, along with the relationship between consciousness and intelligence. Along with superintelligence, it has been long debated whether recreating human consciousness is achievable deemed its complex structures and ambiguous origins. Based on the reviewed research, the findings propose consciousness that mimics human behavior or architecture but does not give rise to phenomenally conscious entities yet. Those types of systems might be beyond our current computational capacities. Finally, we'll discuss some of the moral and ethical consequences that might arise from implementing these types of entities and ultimately caution against the creation of them.

I. INTRODUCTION

The phenomenon of consciousness remains a highly interdisciplinary researched subject by researchers, philosophers, and neuroscientists alike. While its origins are yet to be identified, the need for a greater understanding of consciousness has increased exponentially within the last few years as Artificial Intelligence continues to improve. Some questions surrounding consciousness and AI include, What would it mean for a machine to be sentient? How would we be able to tell if a machine is truly conscious? How would we be able to test for artificial consciousness if the conscious phenomenon is a subjective experience? Attempting to give binary answers to every one of those questions would be premature, thus we will investigate what

research has shown so far involving consciousness and determine if there is enough evidence that suggests the potential of coding true phenomenally conscious entities.

In section 1A, we begin with the various definitions of consciousness and acknowledge the constraints that research in this area is bounded by. In section 1B, we discuss some of the philosophical views of consciousness, along with research that suggests unified sentience. In section 1C, we discuss the concept of qualia, and what artificial qualia might look like, along with potential implementations of it. In section 1D, we'll examine natural vs machine comprehension and what differentiates the two. In section 1E, we'll explore some current tests for consciousness in humans and machines. In section 1F, we'll examine natural vs machine comprehension and what differentiates the two. Lastly, we'll briefly touch upon the ethical implications of a strong artificial consciousness and question the necessity for it. We'll then tie those pieces together to show that a functional artificial consciousness can be implemented but the physiological dimension of consciousness might be beyond our current computing infrastructure at the moment.

A. Definitions of consciousness

To argue for or against the necessity of artificial consciousness, it is important to acknowledge the many pending definitions of consciousness as well as past research that has been done to understand the origins of consciousness. Just a handful of the many emergent definitions of consciousness include:

- "Phenomenal consciousness is experience; what makes a state phenomenally conscious is that there is something "it is like" to be in that state" (Block, 1995, p228)

- "Awareness and wakefulness represent the two main components of consciousness. Awareness is defined by the content of consciousness, and arousal is defined by the level of consciousness. Awareness contains self-awareness, which perceives the internal world of thoughts, reflection, imagination, emotions, and daydreaming, as well as external awareness, which perceives the outside world with the help of the five senses" (Vithoulkas & Muresanu, 2014)

There has also been extensive interdisciplinary research about the origins of consciousness in humans and its functional properties, some of which include:

- Global Workspace Theory that describes a distributed network of neurons, which take in various sensory inputs from across the brain, but come together to create a unified version of reality or consciousness (Baars, 1997).
- Integrated Information Theory which serves as a counter argument to Global Workspace Theory, gives measurement to consciousness through a value called ϕ , which encompasses integration and richness of information (Tononi, 2004).
- Learning Intelligent Distribution Agent (LIDA) is a software model based on the Global Workspace Theory, involving varying neuropsychological theories. It's cognitive cycles are split into 3, understanding, consciousness phase and action phase (Franklin, 2003).

While not having an established definition of consciousness may complicate or potentially undermine certain research when a definition is reached, it should not halt research. Nevertheless, a universal definition might never be attained, in which consciousness will remain a mental phenomenon. Researchers such as David Levy (Levy, 2009) prefer to take this pragmatic approach when working with varying definitions. Despite there being no clear definitions, we can still draw some inferences thus we should devise ways to work around this.

B. The Mind-Body Problem and Consciousness

The Mind Body dilemma was denoted by philosopher René Descartes, which ultimately questions the relationship between the physical body and the mind, where the mind is an intangible, conscious entity. This philosophical approach can usually be divided into 2 main perspectives, the dualism view

which believes the mind and body are two separate entities, as opposed to the monist view which views the mind and body as one entity.

In Roger Sperry's original work on split-brain patients (Sperry, 1973), he portrayed the functional properties of the brain - with right and left hemispheres having differing functions when the corpus callosum was split. For example, if an item was displayed in the left perception field, the patient asserts that they did not see anything, however, the patient can draw the item with their left hand. These observations initially led to the possibility that splitting the brain potentially introduced two different conscious entities (Haan, Corballis Hillyard, 1973).

A research study in 2017 recreated the split brain experiments and challenged this initial ideology (Pinto et al., 1973). Similar to the original research, the corpus callosum was severed and patients were tested over several years. In the experiments, patients were presented with a series of localization and tasks, as well as specifying object orientation and equality of shapes across various places in the visual field. In the end, patients are asked to rate their confidence in their answers to understand if the information they were using to formulate their answers was processed consciously or unconsciously. The experiments were able to replicate the standard findings of the split-brain research, in which visual stimuli cannot be distinguished across left and right visual fields. However, the experiments portrayed a strong correlation between the patients' confidence levels and their accuracy - the patients consistently performed better on the trials they rated with high confidence. Additionally, like the original split-brain work, patients noted no change in their conscious feeling and no other conscious entities present. This research suggests the existence of a unified consciousness and notes that it challenges the Integrated Information Theory (Tononi, 2004) and Global Workspace Theory (Baars, 1997). These frameworks include the assumption that consciousness is also divided in a split brain, which this current research counters.

Having an understanding of the mind-body problem, and how the splitting of the brain might or might not affect consciousness helps us gauge the possibility of implementing consciousness in a non-human entity, which will be discussed in a further section.

C. Consciousness and Artificial Qualia

The term qualia are denoted as "phenomenal properties of experience" (Kind). This is what some researchers define as the "hard problem of consciousness", initially coined by philosopher David Chalmers (Chalmers, 1995), which questions how the processing in our brain gives rise to subjective experiences.

One of the roadblocks towards reaching the concept of phenomenally conscious systems is the implementation of artificial qualia, which by definition would require the hard problem of consciousness to be solved. Researchers Raúl Arrabales, Agapito Ledezma, and Araceli Sanchis address this in their paper, "On the Practical Nature of Artificial Qualia" (Arrabales, Espino Miguel, 2010), where they bring about a potential design for artificial qualia. They allude to some of the prevalent issues when it comes to replicating this human phenomenon, including not having access to first-person perspectives of consciousness. Towards this notion, they argue that while the hard problem of consciousness is irrefutable, perhaps third-person accounts and models can still be utilized to make advances in understanding qualia.

Arrabales, Ledezma, and Sanchis break down the concept of qualia into 3 different stages, which can be used as a potential model for supporting qualia in machines. They label the first stage as "Perceptual Content Representation" where sensory data is consumed and integrated. It is in this stage, that several partial reconstructions of the world contend for a spot or match. This moves onto stage 2, "Introspective Perceptual Meta-Representation", which involves the observation of stage 1, the idea being that if a machine can observe its content creation, it might derive its own ideology about what it's like to have subjective content. The last stage is defined as "Self-Modulation and Reportability", where it utilizes the data developed in the first 2 stages to continue to adjust itself, or a "self-regulation loop". An example of this characterization of artificial qualia that they give is the perception of a moving dot across several images. In stage 1, the machine takes sensory data and identifies that this is a moving dot. In stage 2, it questions what it's like to see a moving dot. Then in the final stage, it reports observing a moving dot.

Recognizing how qualia might be artificially im-

plemented in machines helps us close the gap between concept to reality as well as aid in the understanding of natural qualia. With this research, we have a potential architecture to implement artificial qualia. Similar to the LIDA model, this is based on human cognitive architecture, which furthers its validity. However, it does not account for the subjective experience of consciousness, which suggests that we might be able to mimic general neurological processes of consciousness through code at the moment, but not directly create phenomenally conscious entities.

D. Artificial Consciousness and Comprehension

Another controversial subject of artificial consciousness questions if machines can be sentient if they cannot possess genuine understanding. This dates back to John Searle's famous Chinese Room argument (Searle, 1983), which contends the notion of strong AI, under the impression that machines are simply following a rulebook of information to process inputs and outputs, thus lack acknowledging attributes to true understanding, such as context, models and schemas.

Some theorists argue that computers don't need genuine understanding, as long as they perform well in problem-solving, decision making and language processing (Anderson, 2017).

Perhaps if superintelligence is deemed impossible by some, is it possible to create unconscious superintelligence? Researcher Robert Pepperell discusses the key question in his paper, "Does Machine Understanding Require Consciousness" (Pepperell, 2022). He does so by first determining the key differences between natural, artificial, and machine understanding. Where natural understanding is "The human-like capacity for understanding that is instantiated in our neurobiology, in particular in our brains", artificial understanding is "The capacity for understanding that is implemented in machine learning algorithms as instantiated in digital computers", this can be equated to highly trained classifiers such as AlexNet. Lastly, machine understanding is "The human-like capacity for natural understanding implemented in a non-human mechanical substrate". He then goes on to define some of the intrinsic properties in natural understanding using understanding domain arts as an example. Pepperell ultimately narrows it down to these key properties - learning, recognition, differentiation, integration, context,

reasoning, prediction, consciousness, insight, and reward. This is then compared to artificial understanding, such as image classifiers. These systems perform at high accuracy and at times outperform humans, so it's plausible to argue that they are reaching a conceivable level of understanding. He then goes out to point out that artificial understanding shares the key properties that natural understanding does, except for consciousness, insight, and reward.

Outlining the differences between natural and artificial understanding can help us further recognize the missing pieces towards obtaining machine understanding. As Pepperell argues, consciousness is one of the principle elements that allows humans to perceive and interpret information robustly and in dynamic, ever-changing environments. Thus if consciousness is a pre-requisite for genuine comprehension, and understanding is required for intelligence, then it also seems highly likely that consciousness is required for machine intelligence to match that of humans.

E. Testing for Artificial Consciousness

Currently, there are different ways to test for consciousness in humans. In medical practice, the Grady Coma Scale (Tindall, 1990) has been developed and used frequently to assess consciousness in patients, which breaks down their state of awareness into 5 categories, five being completely unresponsive and one being awake and responsive to stimuli and pain. Having a scale, or test for consciousness helps medical practitioners diagnose and track patients.

The same can be said for machines as well, having no standardized way for testing consciousness continues to blur the definition of what entails conscious machines. That is to say, even if we were to create a machine that is supposedly phenomenally conscious, we would not be able to determine if it was sentient or not. Many of the tests used for humans cannot be directly translated to machines because many of the consciousness tests for humans require embodiment and sensory inputs, which many machines would lack.

There is an abundance of tests that test for artificial consciousness, many either based on or off of Alan Turing's Turing test or claim for sufficient conditions for artificial consciousness but not a necessity. Some examples include:

- Human-Like Conscious Access (Goertzel, 2014), outlines a test based on six key figures of human consciousness - dynamic representation, the focus of energetic resources, the focus of informational resources, global workspace dynamics, integrated information, and correlation of attentional focus with self-modeling.
- Porter Method (Porter, 2016), which bases consciousness on a scale of 0-100 through a series of questions based on the ability to use reason and logic, situational awareness, natural language ability, goals, emotions, options, experiencing existence, self-knowledge, growth, curiosity, and learning. Some sample questions include
 - Can subject identify his/her/its current emotions/feelings/moods?
 - Does subject have a memory of past experiences and events?
 - Does the subject have goals, motivations, and/or needs?

Not having universal consciousness tests for both machines and humans proves to be equally as channeling as not having a definition of consciousness.

F. Implications of Artificial Consciousness

Many negative ramifications can arise if we implement consciousness in machines without discussing its consequences. There is an abundance of ethical issues that arise mostly from how these entities are created and treated by their audience. Some of them are - is it ethical to recreate consciousness, one of the most fundamental properties of humans? Is it ethical to be placing this inside of machines without their consent?

This caution is not new, many researchers argue against building artificial minds including John Basl (Basl, 2013) and Sander Beckers (Beckers, 2018).

Basl denotes the term "moral patient", in which an entity is categorized as one if we need to consider the patient's interests in our deliberations when theirs are at risk. He uses nonhuman entities such as animals to exemplify the concept of a moral patient. Experimenting with them, fighting for entertainment, etc are controversial because they affect their well-being. Basl also argues that performing cognitive research or enhancements on nonhuman entities is potentially more dangerous than typical experiments because of the higher likelihood of

mistreatment; firstly that consciousness might alter systems in indistinguishable ways that the system might not be able to communicate either. Basl suggests some ways to override this belief against the creation of artificial consciousness. One is that if studies should result in the creation of moral patients, those involved will research the needs and interests of said patients, and determine the appropriate treatment for the patient.

Beckers outlines a somewhat similar argument (Beckers, 2018) over the creation of artificial consciousness and outlines it over several key properties. The first is if a super-intelligent agent possesses human traits including suffering, and superintelligence exceeds that of a human, then the notion of super suffering is also viable. This is supported by works of Sotala and Gloor (Sotala, Gloor, 2021), who denotes through several thought experiments that these intelligent agents, "may enable the creation of mental states that are worse than the current bio-psychological limits". One counterargument that Beckers addresses is that perhaps the benefits of greater humanity outweigh the costs of super suffering. He counters this with the point of non-discrimination, that we can't discriminate between the suffering of AI vs the suffering of humans, especially if this suffering might exceed quantities than we can imagine.

Another important ethical aspect of conscious machines is their moral treatment and rights. Levy states that "Consciousness or the lack of it has been cited as the quality that generally determines whether or not a recipient is deserving of ethical treatment" (Levy, 2009). Currently, there are no rules or regulations that enforce the ethical treatment of artificial beings, however, there are many researchers who advocate for the need for legality (Levy, 2009) (Darling, 2016) (Tavani, 2018). Eric Martinez and Christoph Winter's research in "Protecting Sentient Artificial Intelligence: A Survey of Lay Intuitions on Standing, personhood, and General Legal Protection" surveyed over one thousand individuals over their option of giving legal protection and personhood to conscious artificial intelligence compared to different groups such as humans, corporations, animals, and humans in the far and near future. Their findings found that only about a third of individuals wished for personhood for AI, which was the lowest among tested groups. Similarly, the survey also found the desired

level of protection of conscious machines ranked lowest except for corporations. While this survey only accounts for a snapshot of the population and perhaps may evolve, the results are fairly clear - many believe that machines should not have the same moral and legal protections that humans have. However, this perspective might shift over time as society is introduced to stronger variants of artificial intelligence in their daily lives. It wasn't too long ago that various groups were considered legal non-persons and deprived of basic human rights. Some equate the creation of conscious machines with the sole purpose to serve their creators as a form of servitude (Edelman, 2021).

II. ANALYSIS

In the prior section, several literature pieces and research were presented that discussed the complex and emerging properties of recreating consciousness, including phenomenology and artificial qualia, machine understanding, and machine tests for consciousness. These properties are some of the stronger arguments that most would use to refute the possibility of seemingly doing the impossible: equating algorithms and machine processing to the currently enigmatic phenomenon of consciousness.

In Pinto's 2017 split brain study, it found to be that a split-brain did not give rise to split consciousness. This results in several important elements, firstly it supports the theory of dualism, that the mind and brain are separate entities, refuting the famous Global Workspace Theory. If this supports dualism, then this implies consciousness as a non-physical property and something outside the brain, making it difficult to argue that consciousness is something we will be able to code.

In Raul Arrabales, Agapito Ledezma, and Araceli Sanchis's research, we see a potential architecture for modeling artificial qualia. However, it only mimics the flow of human cognitive processes but does not give strong evidence towards phenomenally conscious entities as it avoids the subjective experience.

In Pepperell's research, we see that consciousness is one of the key properties that differentiate natural/human understanding from machine understanding. Thus, under this framework, solving Searle's Chinese room argument would require a working model of a phenomenally conscious agent to inspire true comprehension of the material.

The reviewed research delved into commonly debated sub-topics within artificial consciences research. While many are extremely promising, there is still no definitive proof or modeling that would give rise to phenomenally coded consciousness. While there is still much left to be said and further researched about the state of consciousness both in humans and machines, it's perhaps premature to deny its future existence in systems. Some recent research in neuroscience this year has suggested that brain functions might be operating on a quantum level. Dr. Kerskens and his team at Trinity College London have potentially found evidence of proton entanglement in the brain through MRIs. Entanglement in the brain opens the possibility that the brain functions might be quantum (Kerskens, López Pérez, 2022). There are a number of researchers that also question and give evidence to support the probability of quantum cognition (Fisher, 2015) (Stuart, 1998) . Thus it might be plausible that consciousness can be encoded, but it's beyond our current computing infrastructure.

III. CONCLUSION

This article reviewed some recent research that endorses the potential of conscious machines. Overall these endorse the possibility of creating machines that mimic the behaviors of conscious entities, however creating the physiological dimension of consciousness might be beyond our current capabilities.

Nevertheless, as we visit the concept of moral persons and the evolving dynamic between humans and autonomous agents, we contend that the implications of creating conscious entities might exceed its necessity. The lack of rules and regulations around conscious entities and their treatment of these beings makes this type of research very precarious. As this interdisciplinary field continues, there should be greater oversight to consider the appropriate treatment of these created entities and how they might fit into our societal structure.

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